

SkateFormer Person-Track Continuity

Formal Experimental Report

NTU RGB+D 120 · Interaction-26 · Cross-Subject
Global, Temporal, Localized and Sliding Localized Person Swap

11,660

Test samples

26

Two-person interaction classes

64 frames

Sampled model input

Principal findings

The trained SkateFormer pipeline is highly robust to whole-clip person-slot exchange but remains strongly vulnerable when person tracks are broken within a sequence. Fixed-duration localized interventions show that temporal position matters: later sampled-input regions are generally more sensitive, although the response is not strictly monotonic.

−0.15 pp

Fusion Top-1 under Global Swap

−22.06 pp

Fusion Top-1 at W10 [40,56)

28.06%

Fusion prediction flips at W10

Evidence-led conclusions

- Global robustness is consistent with verified whole-clip person-swap augmentation, but cannot be attributed solely to architecture.
- Localized Early/Middle/Late Fusion degradation is −8.03/−14.96/−22.06 pp.
- Fusion accuracy and flip rate peak at center 48, while Fusion mean TV peaks earlier at center 40.
- The primary improvement target is temporal person preservation and pair-relation modeling, not additional global permutation invariance.

Scientific scope. The experiments diagnose sensitivity to person-track continuity disruption. They do not demonstrate human identity recognition, and the temporal interventions can introduce artificial spatial, trajectory, velocity and motion discontinuities.

Research question and experimental scope

Two-person skeleton action recognition receives an explicit two-slot representation. The central question is whether predictions depend on arbitrary slot order or on maintaining a consistent assignment of each observed skeleton to a slot over time.

Preserved by the intervention

- The two skeletons present at every frame.
- Coordinates, frame order, labels and sample identity.
- The unordered per-frame pair of people.
- The trained checkpoint and test pipeline.

Changed by the intervention

- The assignment of skeletons to Person-0 / Person-1 slots.
- For partial swaps, continuity of each slot through time.
- Potentially, trajectory and velocity continuity at swap boundaries.

Why accuracy alone is insufficient

Aggregate accuracy can hide substantial output instability: correct→wrong and wrong→correct transitions can partially cancel. The study therefore reports Top-1, Top-5, their percentage-point changes, prediction flip rate, total variation (TV) between clean and perturbed probability distributions, and four correctness transitions.

Temporal coordinate. A window location refers to the sampled 64-frame model input after deterministic test preprocessing. It is not necessarily the same absolute time in every original NTU video.

SkateFormer baseline

The diagnostic experiments use the completed two-stream baseline without retraining. Model input is **[B, 3, 64, 24, 2]**. Joint and Bone were trained independently; Fusion is a fixed score operation, not a learned third model.

Output	Selected epoch	Test Top-1	Test Top-5	Definition
Joint	439	91.24%	98.37%	Coordinates
Bone	497	89.90%	97.80%	Runtime bone vectors
Fusion	—	92.20%	98.88%	$S_{\text{fusion}} = 0.5 S_{\text{joint}} + 0.5 S_{\text{bone}}$ (fixed raw-logit fusion)

11,764

Train samples

1,308

Validation samples

11,660

Test samples

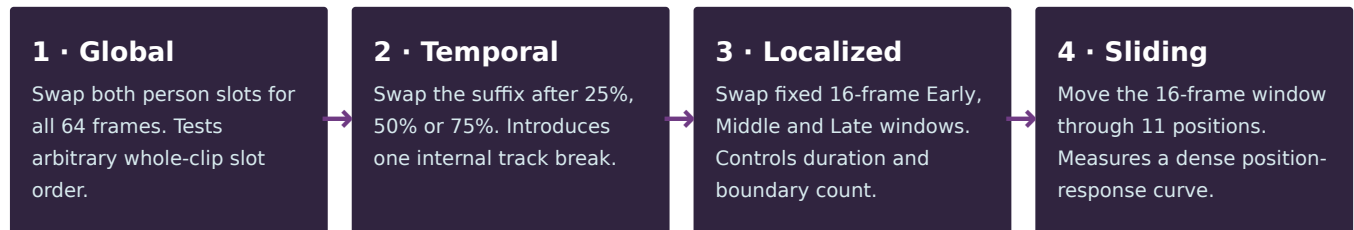
Test protocol

- Dataset: NTU RGB+D 120, Interaction-26 subset, Cross-Subject split.
- Deterministic test input length: 64 frames; two person slots.
- Checkpoint selection was completed before diagnostic test analysis.
- All four diagnostics reuse the same clean baseline scores and class labels.

Training observation. Whole-clip person-slot swapping was used during training through the verified augmentation configuration. This observation is relevant context, not proof of the mechanism behind the measured response.

Four-stage experimental progression

The sequence was designed so that each stage addresses an ambiguity exposed by the preceding one.



Controlled semantics

- Global: all frames, no internal boundary.
- Temporal: variable swapped duration and one internal boundary.
- Localized/Sliding: 16 frames and exactly two internal boundaries.

- No checkpoint or fusion-weight search.
- No test-time randomness.
- No model retraining.
- Predictions, distributions and transitions independently validated from saved artifacts.

Interpretive logic

Global Swap separates whole-clip slot order from temporal track continuity. Temporal Swap establishes sensitivity to within-sequence disruption but confounds breakpoint and duration. Fixed Localized Swap removes that confound for three positions; Sliding Localized Swap tests whether the three-point pattern persists across the sampled timeline.

Global Person Swap

For every sample, the model-input person axis is exchanged for the entire clip: Person-0 ↔ Person-1 at every frame. Labels and the per-frame skeleton pair remain unchanged.

Stream	Condition	T1	Pert. T1	$\Delta T1$ pp	$\Delta T5$ pp	Flip	Mean TV
Joint	Global	91.24%	91.23%	-0.02	-0.01	1.67%	0.0173
Bone	Global	89.90%	89.78%	-0.12	0.10	2.11%	0.0214
Fusion	Global	92.20%	92.05%	-0.15	-0.05	1.22%	0.0151

Observed response

All outputs are stable under whole-clip exchange, especially Fusion (-0.15 pp Top-1; 1.22% flips). Non-zero flips and TV show that the model is not mathematically invariant. Verified training-time whole-clip swap augmentation is a major interpretive caveat.

Probability-scale note

Fusion probabilities are computed from the fixed $0.5J+0.5B$ raw-logit rule. No fusion weight was selected on the test set.

Fusion condition	C→C	C→W	W→C	W→W
Global	10,687	64	46	863

C→C: clean correct and perturbed correct; C→W: clean correct and perturbed wrong; W→C: clean wrong and perturbed correct; W→W: wrong in both conditions. Counts sum to 11,660.

Fixed-Breakpoint Temporal Swap

Frames before the breakpoint retain their slot assignment; all later frames exchange the two slots. The three conditions create one internal track discontinuity but differ in swapped duration.

Stream	Condition	T1	Pert. T1	$\Delta T1$ pp	$\Delta T5$ pp	Flip	Mean TV
Joint	25% / f16	91.24%	80.70%	-10.54	-4.81	15.85%	0.1629
Bone	25% / f16	89.90%	85.57%	-4.33	-2.13	9.00%	0.0940
Fusion	25% / f16	92.20%	86.18%	-6.02	-2.87	10.03%	0.1161
Joint	50% / f32	91.24%	78.10%	-13.15	-4.66	19.28%	0.1939
Bone	50% / f32	89.90%	79.76%	-10.14	-2.38	16.54%	0.1648
Fusion	50% / f32	92.20%	82.89%	-9.31	-2.40	14.31%	0.1627
Joint	75% / f48	91.24%	77.57%	-13.67	-4.54	20.50%	0.2107
Bone	75% / f48	89.90%	77.18%	-12.72	-3.93	19.98%	0.1894
Fusion	75% / f48	92.20%	80.41%	-11.79	-3.10	17.52%	0.1860

Interpretation. Fusion degradation increases from -6.02 to -9.31 to -11.79 pp as the breakpoint moves later, even though fewer suffix frames are swapped. This suggests a position effect, but duration and position remain confounded.

All temporal conditions produce a larger Fusion response than Global Swap. This establishes sensitivity to within-sequence track disruption and motivates the fixed-duration Localized experiment.

Fixed 16-frame Localized Swap

Early [8,24), Middle [24,40) and Late [40,56) each exchange exactly 16 frames and introduce exactly two internal boundaries. Only temporal position changes.

Stream	Condition	T1	Pert. T1	$\Delta T1$ pp	$\Delta T5$ pp	Flip	Mean TV
Joint	Early [8,24)	91.24%	80.13%	-11.11	-4.97	16.72%	0.1752
Joint	Middle [24,40)	91.24%	75.27%	-15.97	-8.17	22.52%	0.2376
Joint	Late [40,56)	91.24%	71.96%	-19.29	-6.68	26.56%	0.2629
Bone	Early [8,24)	89.90%	82.32%	-7.57	-3.22	13.32%	0.1376
Bone	Middle [24,40)	89.90%	70.83%	-19.07	-6.48	26.38%	0.2530
Bone	Late [40,56)	89.90%	64.31%	-25.59	-11.38	33.25%	0.3023
Fusion	Early [8,24)	92.20%	84.18%	-8.03	-3.36	12.58%	0.1437
Fusion	Middle [24,40)	92.20%	77.25%	-14.96	-5.86	20.58%	0.2296
Fusion	Late [40,56)	92.20%	70.15%	-22.06	-6.20	28.06%	0.2725

-8.03 pp

Fusion Early $\Delta T_{\text{Top-1}}$

-14.96 pp

Fusion Middle $\Delta T_{\text{Top-1}}$

-22.06 pp

Fusion Late $\Delta T_{\text{Top-1}}$

Late > Middle > Early for Fusion degradation, flip rate and mean TV. Because duration and boundary count are fixed, the result supports a genuine temporal-position dependence within the sampled model input. It also resolves the ambiguity of the suffix experiment: its apparent peak cannot be read as an intrinsic middle-region maximum.

The intervention remains artificial. Equal duration does not make the swap a pure identity manipulation, and it does not imply that window length is irrelevant in other settings.

Sliding Localized Swap: 11-point response

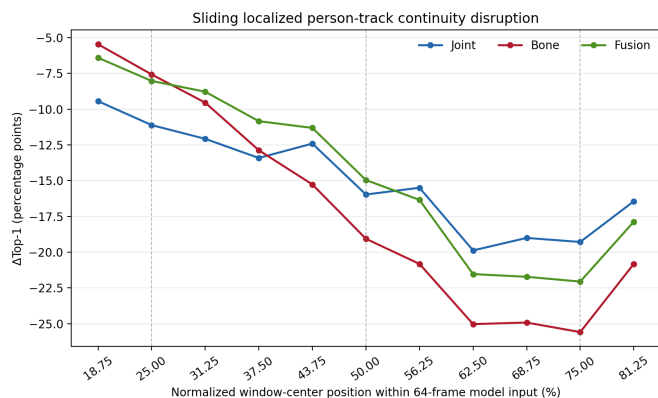


Figure 1. Top-1 degradation across window centers.

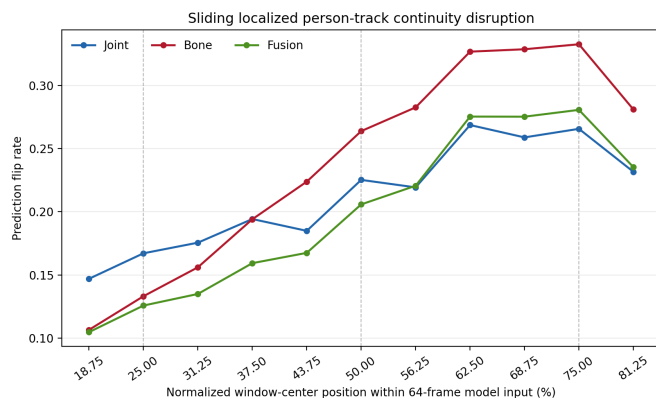


Figure 2. Prediction flip rate.

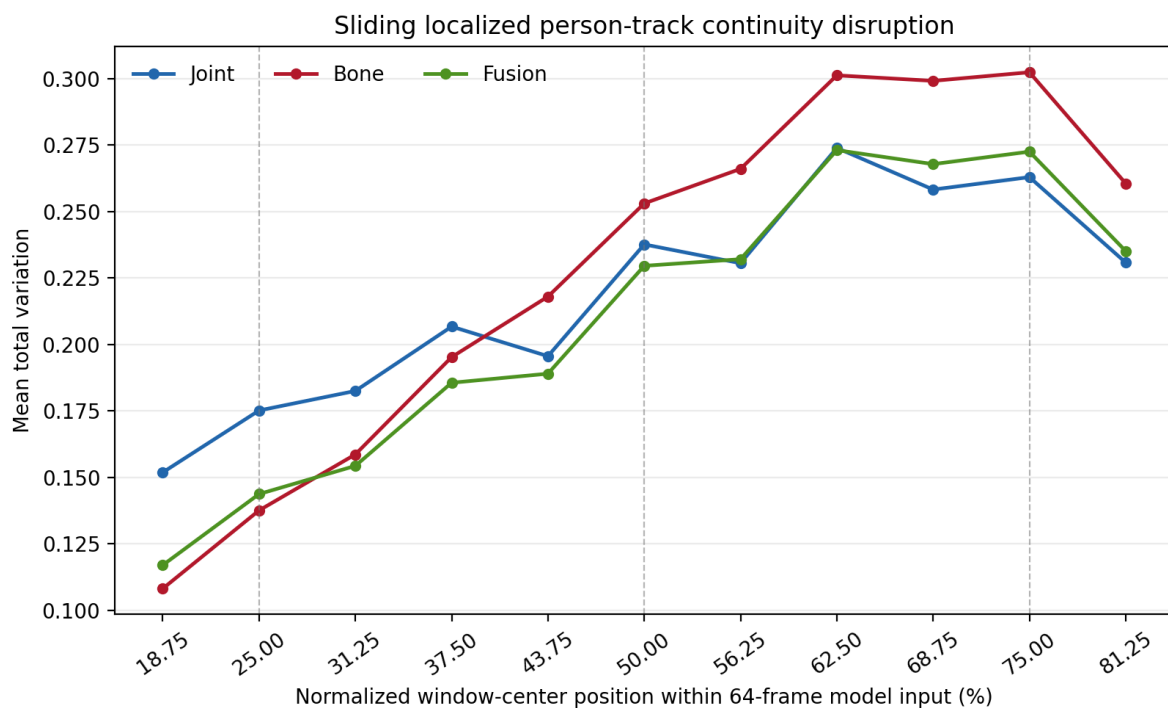


Figure 3. Mean total variation. For CTR-GCN, Fusion uses normalized 0.5J+0.5B probabilities; Joint and Bone use their native logits.

The curves rise broadly toward the mid-late and late sampled-input regions, but are not strictly monotonic. W02, W06 and W10 reproduce the earlier Early/Middle/Late checkpoints; all three regression checks passed.

Joint, Bone and Fusion response

Stream	Early avg $ \Delta T1 $	Midpoint	Late avg $ \Delta T1 $	Max Δ window	Max flip	Max TV
Joint	11.69	15.97	18.02	W08 / c40	26.86%	0.2739
Bone	10.14	19.07	23.44	W10 / c48	33.25%	0.3023
Fusion	9.07	14.96	19.91	W10 / c48	28.06%	0.2731

Joint accuracy degradation peaks at center 40; Bone peaks at center 48. Fusion Δ Top-1 and flip rate peak at center 48, while Fusion mean TV peaks at center 40. Fusion improves clean performance and usually moderates the more vulnerable stream, indicating complementary responses without proving a causal mechanism.

Fusion 11-point values

ID	Window	Center	$\Delta T1$ pp	Flip	Mean TV
W01	[4,20)	12	-6.41	10.48%	0.1170
W02	[8,24)	16	-8.03	12.58%	0.1437
W03	[12,28)	20	-8.78	13.50%	0.1543
W04	[16,32)	24	-10.84	15.93%	0.1856
W05	[20,36)	28	-11.31	16.75%	0.1890
W06	[24,40)	32	-14.96	20.58%	0.2296
W07	[28,44)	36	-16.35	22.06%	0.2321
W08	[32,48)	40	-21.54	27.53%	0.2731
W09	[36,52)	44	-21.72	27.52%	0.2678
W10	[40,56)	48	-22.06	28.06%	0.2725
W11	[44,60)	52	-17.87	23.53%	0.2351

Early average uses centers <32 ; midpoint is center 32; late average uses centers >32 . Spearman position trends are exploratory summaries over only 11 measured positions, not causal tests.

Where sensitivity concentrates

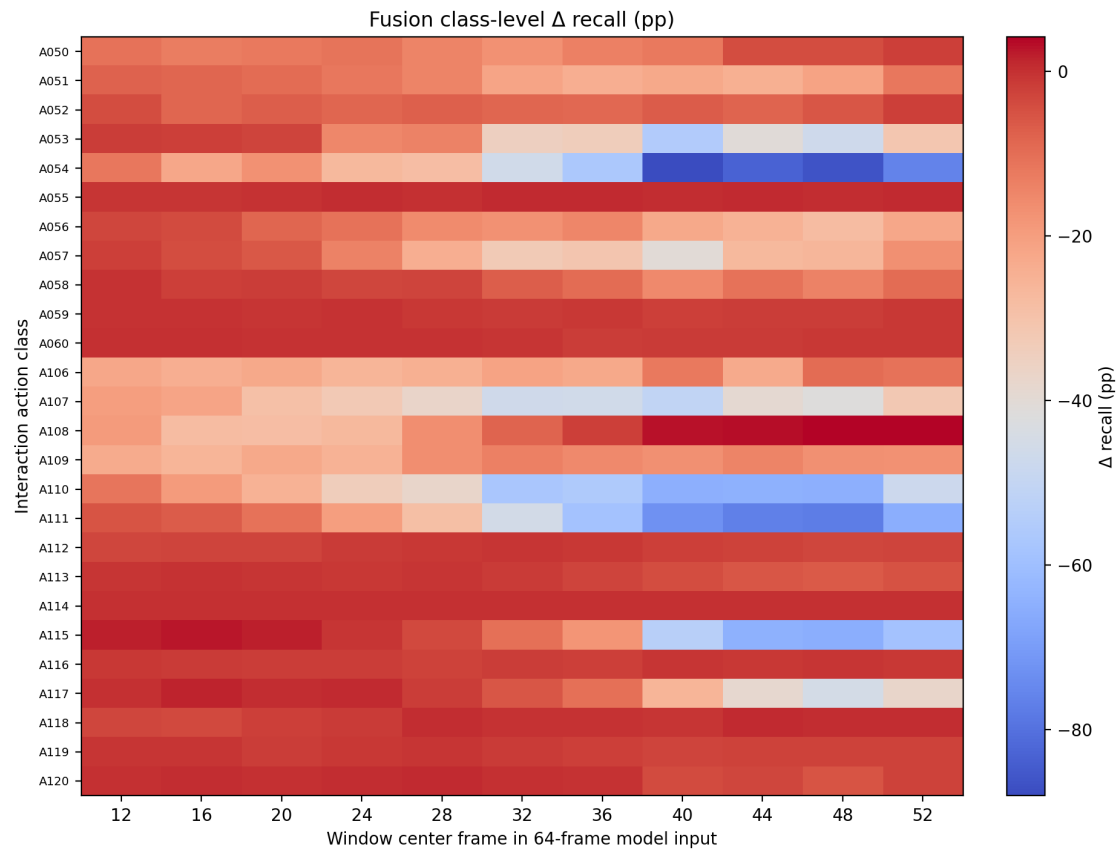


Figure 4. Fusion recall change by class and sliding window. The complete 26×11 values remain in the experiment artifacts; the report presents only selected summaries.

Largest late-minus-early concentration

Class	Late-early drop	W10 drop	W10 flip
A054 — point finger at the other person	+56.74 pp	86.59 pp	90.58%
A111 — step on foot	+55.69 pp	77.57 pp	81.74%
A115 — take a photo of other person	+49.86 pp	65.10 pp	68.92%
A053 — pat on back of other person	+34.20 pp	46.74 pp	48.91%
A110 — shoot at other person with a gun	+33.98 pp	64.70 pp	80.70%

Observed failure patterns

- Whole-clip slot-order sensitivity: small across all outputs.
- Strong within-sequence track-disruption sensitivity.
- Temporal-position-dependent vulnerability, concentrated later.
- Different Joint and Bone response shapes.
- Substantial class-dependent variation.

Relatively stable on mean recall drop: A114 — carry something with other person, A055 — hugging other person, A060 — walking apart from each other, A059 — walking towards each other.

The action IDs and names shown are taken from the verified SkateFormer per-class artifacts. Class rankings describe concentration of measured response; they do not identify a semantic cause.

Evidence-to-design reasoning

Experimental evidence	Observed failure pattern	Design requirement	Candidate modification	Support / complexity / risk
Global Fusion: only -0.15 pp	Global order is not the primary failure	Do not prioritise extra invariance	Retain the verified augmentation; focus elsewhere	High / none / low
Large Localized/Sliding degradation	Person tracks are lost within sequences	Preserve paired temporal features	Keep $h_A(t)$, $h_B(t)$ until late aggregation	High / medium / medium
Late/mid-late sensitivity	Temporal evidence is concentrated	Broader interaction-time coverage	Multi-scale or phase-aware interaction attention	High / medium / medium
Joint/Bone peaks differ	Modalities carry complementary response	Earlier controlled cross-stream exchange	Small relation fusion block before classification	Medium / medium / medium

Observation Measured global, temporal and class-dependent response	Design implication Preserve symmetry or temporal pairing where the evidence demands it	Minimal modification Change one representation or loss component and ablate it independently
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Priority. The first intervention should preserve paired-person features and add a lightweight symmetric relation representation before final aggregation. Additional global swap regularisation is a lower priority because the trained pipeline is already robust to whole-clip exchange.

Explicit cross-person relations—relative joint displacement, distances, relative velocity, relative bone geometry or learned pair tokens—are plausible later directions. They remain hypotheses until clean and diagnostic ablations show a benefit.

Lightweight symmetric paired-person relation block

Retain separate person features through the final temporal representation, form a symmetric pair descriptor such as $[h_A + h_B, |h_A - h_B|]$, project it through one small layer, and add it residually before classification. Parameter overhead is conceptually $O(d^2)$ for one projection and should be reported exactly after implementation.

Required ablation

Baseline; delayed person aggregation only; symmetric relation descriptor only; combined block. Keep the existing whole-clip swap augmentation and fixed Joint/Bone fusion unchanged.

Baseline comparison

Use the same data split, checkpoint-selection procedure and clean evaluation. Do not choose the modification from Test perturbation results.

Success criteria

- Maintain or improve clean Test Top-1; report Macro-F1 when implemented.
- Report per-class recall, parameter count and computational overhead.
- Reduce the model-relevant Global and selected Localized/Sliding sensitivity.
- Verify that robustness is not obtained by suppressing useful motion evidence.

Limitations

The study uses one trained baseline line and artificial slot interventions. It cannot separate identity, spatial jumps, trajectory changes, velocity discontinuities and other motion inconsistencies. Temporal positions are defined only in the sampled 64-frame input. Exploratory position trends and class rankings require confirmation on additional seeds and, ideally, other protocols.

Conclusion

SkateFormer is robust to whole-clip slot order but strongly dependent on consistent within-sequence person tracks. The controlled Localized and Sliding experiments identify a later-region vulnerability without establishing a causal mechanism. The recommended next step is deliberately minimal, testable and reversible.